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Gastrointestinal symptoms correlate with core clinical features and systemic inflammation in myalgic encephalomyelitis/chronic fatigue syndrome

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Abstract

Background Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a debilitating multisystem illness marked by fatigue, cognitive impairment, and post-exertional malaise. Gastrointestinal (GI) symptoms are frequently reported, yet their relationship to central features of the illness and biological correlates remains poorly understood.

Objectives We aimed to characterize GI symptom burden in ME/CFS and evaluate its associations with core clinical features and specific immune and inflammatory markers, with attention to potential gut-related contributions to disease expression.

Methods GI symptoms and 49 additional symptoms across nine domains were assessed in 116 ME/CFS patients and 80 matched controls. Plasma C-reactive protein (CRP) and antibodies against dietary and microbial antigens were measured as indicators of systemic inflammation and putative gut-derived antigen exposure.

Results ME/CFS patients reported significantly elevated GI symptom frequency and severity compared with controls, with 53% of ME/CFS patients versus 8% of controls reporting a prior diagnosis of irritable bowel syndrome. GI symptom burden correlated with fatigue, cognitive difficulties, flu-like symptoms, pain, sleep disturbances, neurological complaints, and sensory sensitivities, independent of illness duration. CRP levels were higher in patients with greater GI symptoms and correlated with GI, fatigue, musculoskeletal pain, and flu-like symptom burden. Patients with greater flu-like symptom expression exhibited higher IgM responses to dietary gliadin and bacterial lipopolysaccharide. These associations were not detected in controls.

Conclusions GI symptoms are a prominent, clinically relevant dimension of ME/CFS, associated with broader symptom burden and inflammatory heterogeneity. These findings highlight the relevance of gut-related and immune processes in ME/CFS and underscore the value of incorporating GI symptom assessment in translational studies to help refine mechanistic understanding and improve therapeutic stratification.

Keywords Myalgic encephalomyelitis/chronic fatigue syndrome, Gastrointestinal system, Immune response, Inflammation, Microbial translocation, Gut permeability, CRP

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Introduction

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a complex, debilitating illness characterized by unexplained and prolonged fatigue that does not improve with rest [1]. As a multisystem condition, ME/CFS often presents with a wide range of symptoms in addition to physical fatigue such as cognitive difficulties, orthostatic intolerance, sleep disturbances, flu-like symptoms, autonomic dysfunction, and post-exertional malaise. The condition disproportionately affects females and it is estimated that roughly 75% of patients are unable to work [2, 3]. The etiology of ME/CFS remains unclear, and its underlying pathophysiological mechanisms are not yet well understood. Diagnosis of ME/CFS can be challenging, which may be attributable to the heterogeneity of clinical presentations, variability of ME/CFS case definitions, and limited understanding of the condition [4–6]. Furthermore, there are currently no validated diagnostic biomarkers or FDA-approved treatments for ME/CFS.

In recent years, increasing attention has been given to immune dysregulation as a key component of ME/CFS [7]. Additionally, individuals with ME/CFS have been found to report a high burden of gastrointestinal (GI) symptoms and irritable bowel syndrome (IBS) has been identified as a comorbidity [8–16], pointing to a potentially important role for GI involvement in ME/CFS. However, the specific nature, frequency, and clinical significance of these GI symptoms have not been defined, and their relationship to core ME/CFS features such as fatigue and cognitive difficulties, or their relevance in the context of the reported immunologic changes, have not been explored. As a result, GI symptoms are often treated as peripheral rather than an integral component of the illness.

In parallel with clinical reports, mechanistic studies have begun to delineate biological pathways potentially linking GI abnormalities to ME/CFS. Alterations in gut microbial composition, metabolic dysfunction, and evidence suggestive of altered intestinal barrier function and microbial translocation have been documented in affected individuals [17–21]. Our prior work identified findings consistent with gut epithelial cell injury and increased exposure to gut-derived immunogenic factors, including elevated circulating LPS levels and enhanced systemic antibody responses to microbial and dietary antigens in ME/CFS [20]. Together, these observations suggest a biologically plausible pathway linking altered mucosal barrier function and gut-associated antigen exposure to systemic immune activation and multisystem symptom expression, including flu-like features, fatigue, and cognitive deficits.

Evidence for low-grade systemic inflammation in ME/CFS, however, has been heterogeneous. For example, while some studies have reported elevated C-reactive

protein (CRP) levels, other well-controlled investigations, including our own rigorously matched cohort study [22] and additional clinical cohorts [23] that likewise observed no significant CRP elevation relative to controls, suggest that CRP is not uniformly increased across patients and may instead characterize inflammatory subphenotypes within the condition. In this context, CRP provides a pragmatic index of systemic inflammatory activity that may be most informative for subgroup analyses in ME/CFS, particularly when considered alongside clinical and other immunological measures. Complementary assessment of antigen-specific antibody responses may provide an additional indirect window into gut barrier dysfunction and microbial translocation, as these processes have been linked to systemic immune activation and flu-like symptomatology in prior studies [24, 25].

Taken together, converging evidence of gut barrier dysfunction, immune activation, and clinical heterogeneity in ME/CFS underscores a critical gap in the literature, as no prior studies have integrated GI symptom profiling with immune biomarkers and hallmark clinical domains within a single analytic framework. To address this gap, we characterized GI symptom burden in individuals with ME/CFS and examined its relationships with core symptom domains alongside circulating markers selected to capture relevant and complementary aspects of immune activity, systemic inflammatory tone (CRP), and potential gut-derived antigen exposure (antibodies to dietary gliadin and bacterial LPS). By integrating GI symptom assessment with analysis of hallmark ME/CFS symptom domains and immune biomarkers, the study evaluates whether GI symptom burden constitutes a clinically relevant axis of disease expression that aligns with core symptom profile and gut-relevant inflammatory signaling.

Methods

Study participants

The study's cohorts included 116 individuals with ME/CFS and 80 healthy control participants. These individuals were recruited under institutional review board protocols as part of a collaboration between Solve ME/CFS Initiative, a group of ME/CFS clinical sites, and Glaxo-SmithKline, as previously described in detail, and were included in the present analyses from the larger repository cohort based on the availability of both stored biospecimens and complete clinical and symptom data [26]. Subjects with ME/CFS were required to meet both the Fukuda and Canadian diagnostic criteria for inclusion [27, 28], along with having an initial presentation of flu-like illness with an acute (48 h) or subacute (4 weeks) onset, fatigue that has persisted for at least six months, post-exertional malaise lasting >24 hours, and significant cognitive impairment in short-term memory and

concentration. ME/CFS participants were also required to satisfy the RAND-36 quality of life survey by meeting 2 of the 3 following benchmarks: vitality <35, social functioning <62.5, and role-physical <50. Healthy control subjects resided within the same neighborhood, zip code or neighboring zip code as the ME/CFS subjects, but did not reside in the same household at the time of recruitment. These individuals were also screened to ensure they did not meet ME/CFS case definition criteria. Participants were excluded if they had a body mass index (BMI) >40 kg/m², cancer, a history of substance or alcohol abuse <2 years before the onset of ME/CFS, untreated hypothyroidism, or a major psychiatric disorder. Study subjects with a history of liver disease were excluded, in addition to individuals with abnormal liver function blood test results for aspartate transaminase (AST), alanine transaminase (ALT), and alkaline phosphatase (ALP). Participants with abnormal levels of total protein, albumin, globulin, and bilirubin were also excluded. Individuals with a chronic inflammatory disease, including rheumatoid arthritis and systemic lupus erythematosus in addition to those reporting a recent infection, were not included. Female subjects who were pregnant, less than 3 months post-partum, or currently lactating were excluded. None of the study participants had a prior diagnosis of inflammatory bowel disease (IBD) or celiac disease.

Blood plasma samples were collected with written informed consent under institutional review board protocols. The samples were kept at -80 °C until use for assays to maintain stability. This study was approved by the institutional review board of Columbia University Medical Center.

Symptom assessment

Study participants had completed questionnaires using standardized 5-point scales assessing the frequency and severity of 54 symptoms relevant to ME/CFS [29]. Frequency was rated as 1 (none of the time), 2 (a little of the time), 3 (about half the time), 4 (most of the time), or 5 (all of the time). Severity was rated as 1 (not present), 2 (mild), 3 (moderate), 4 (severe), or 5 (very severe).

Five GI symptoms were specifically evaluated: bloating, abdominal/stomach pain, nausea, loss of appetite, and bowel problems. Participants also reported whether they had previously received a diagnosis of irritable bowel syndrome (IBS) from a healthcare provider. In addition to the five GI symptoms, the remaining 49 symptoms were grouped into eight other domains representing key aspects of ME/CFS presentation: physical fatigue, cognitive difficulties and mental fatigue, musculoskeletal pain, sleep disturbances, flu-like symptoms, neurological symptoms, sensory sensitivities, and cardiovascular symptoms. Symptom domains were constructed as

clinically oriented groupings based on commonly recognized symptom dimensions in ME/CFS. Symptom items were assigned exclusively to individual domains, with the only exception being “pain/aching in muscles,” which was included in both the musculoskeletal pain and flu-like symptom domains. The individual symptoms comprising each domain are listed in Table 1.

Symptom frequency was used primarily to assess symptom presence and for patient stratification, as it captures how consistently symptoms occur. The presence of individual GI symptoms was defined as a frequency score ≥2 (i.e., “a little of the time” or more). Participants were also stratified into high versus low GI symptom frequency groups based on summed GI frequency scores, using a cutoff of 10 out of 25, corresponding to reporting most of the five GI symptoms as at least “a little of the time”. Similarly, for flu-like symptoms, a common feature of ME/CFS [30], participants were stratified using a cutoff of 13 out of 30.

For each participant, frequency and severity ratings were summed to calculate domain-specific symptom burden scores, reflecting overall symptom impact. These composite scores were used in correlation analyses with other symptom domains and biological markers to capture the combined influence of symptom frequency and severity within a single quantitative measure.

Assays

Plasma levels of CRP, a prototypical acute-phase protein and highly sensitive marker of inflammation [31], were measured by the enzyme-linked immunosorbent assay (ELISA) according to the assay manufacturer’s protocol (Alpco), as we have previously described [22, 32].

Plasma antibodies to selected dietary and microbial components were measured by ELISA, as indirect markers of gut-derived antigen exposure and associated immune responses, consistent with established approaches in other studies [33–35]. The gliadin protein fraction of dietary wheat was used as a representative dietary antigen because of its established immunogenicity, and IgG, IgA, and IgM anti-gliadin antibodies were quantified using a validated in-house assay, as we have previously described in detail [35–37]. Lipopolysaccharide (LPS), a major component of the outer membrane of Gram-negative bacteria, served as the representative microbial antigen, and IgG, IgA, and IgM anti-LPS antibodies were assessed using commercial endotoxin-core antibody (EndoCAB) kits, according to manufacturer’s protocols (Hycult Biotech) [35].

To rule out a potential association between anti-gliadin antibodies and celiac disease, an autoimmune enteropathy characterized by immunologic reactivity to gliadin and related proteins, sensitive and specific markers of celiac disease were assessed. These included anti-tissue

Table 1 ME/CFS symptom domains and their constituent symptoms

ME/CFS Symptom Category	Symptoms Included*
Gastrointestinal Symptoms	Bloating Abdominal/stomach pain Nausea Loss of appetite Bowel problems
Physical Fatigue	Fatigue/extreme tiredness Heavy feeling after starting to exercise Next day soreness or fatigue after non-strenuous everyday activities Physical exhaustion after minimal exercise Physically drained or sick after mild activity Muscle weakness
Cognitive Difficulties & Mental Fatigue	Mentally tired after the slightest effort Problems with memory Difficulty paying attention Difficulty finding the right words or expressing thoughts Difficulty with understanding Only able to focus on one thing at a time Unable to focus vision and attention Slowness of thought Absent-mindedness or forgetfulness
Musculoskeletal Pain	Pain/aching in muscles Pain/stiffness/tenderness in > 1 joint without swelling or redness
Sleep Disturbances	Need to nap daily Problems falling asleep Problems staying asleep Waking up early in the morning (e.g., 3am) Feeling unrefreshed after waking up in the morning Sleep during day and awake at night
Flu-Like Symptoms	Fever Chills/shivers Pain/aching in muscles Sore throat Tender/sore lymph nodes Perceived feelings of flu illness
Neurological Symptoms	Headache Eye pain Muscle twitches
Sensory Sensitivities	Sensitivity to noise Sensitivity to bright lights Some smells, foods, medications, or chemicals induce feelings of sickness
Cardiovascular Symptoms	Chest pain Dizziness or fainting Irregular heartbeat

*Symptom items were assigned exclusively to individual domains, with the only exception being “pain/aching in muscles,” which was included in both the Musculoskeletal Pain and the Flu-Like Symptoms domains

transglutaminase (anti-TG2) antibodies, anti-deamidated gliadin antibodies, and celiac disease associated HLA genes. Plasma IgA antibody to recombinant human TG2 was measured by ELISA, according to the manufacturer’s protocol (Euroimmun AG). IgA and IgG antibodies to a previously described glutamine-glutamate substituted

trimer of a fusion peptide containing the sequences PLQPEQFPF and PEQLPQFEE [38], representing deamidated gliadin, were measured by separate ELISAs, according to the manufacturer’s protocols (Euroimmun AG). High-resolution HLA-DQA1 and -DQB1 genotyping was performed as we have previously described [39], using multiplex PCR followed by sequence-specific oligonucleotide (SSO) probe detection [40]. Amplified DNA was hybridized to fluorescent bead-based probes targeting DQA1 and DQB1 polymorphic sequences (One Lambda) and analyzed on a LABScan 100 platform using Luminex technology with manufacturer-provided interpretation software. The presence or absence of celiac disease-associated alleles DQA10501/0505-DQB10201/0202 (DQ2) and DQA103–DQB10302 (DQ8) was determined.

Data analysis
Demographic differences between cohorts were assessed using Fisher’s exact test for categorical variables (sex and race) and unpaired two-tailed t-tests for continuous variables (age and BMI). Continuous variables were screened for normality of distribution, equality of variances, and outliers prior to comparative analyses. For normally distributed data, between-group comparisons were performed using unpaired two-tailed t-tests; Welch’s correction was applied when variances were unequal. Variables that deviated from normality were log-transformed prior to analysis.

Correlations between GI symptoms, predefined ME/CFS symptom domains, and biological markers were examined using partial correlation analyses implemented through linear regression modeling with age, sex, and BMI included as covariates. For variables that remained non-normally distributed after transformation, rank-based (Spearman) partial correlations were performed following covariate adjustment using regression residuals.

All statistical tests were two-sided, and $p < 0.05$ was considered statistically significant. Analyses were conducted using Prism 10 (GraphPad Software). Primary analyses were pre-specified based on prior biological and clinical observations in ME/CFS. To account for multiple testing within related multidomain analytical families, false discovery rate correction was performed using the Benjamini-Hochberg procedure for correlations between GI symptom burden and predefined ME/CFS symptom domains, multidomain symptom-correlation analyses involving CRP, and multidomain analyses involving antigen-specific antibody responses.

Results
Study participants

The demographic characteristics for the study cohorts are presented in Table 2. ME/CFS and healthy control

Table 2 Demographic and clinical characteristics of ME/CFS and healthy control cohorts

Variable	ME/CFS Patients (n = 116)	Healthy Controls (n = 80)	P value*
Age—years [SD]	49.7 [11.9]	49.6 [13.2]	ns
Female sex—no. (%)	80 (69.0)	64 (80.0)	ns
White race—no. (%)	112 (96.6)	78 (97.5)	ns
BMI—(kg/m ²) [SD]	26.0 [5.6]	26.5 [6.8]	ns
Duration of illness—years [SD]	16.3 [9.9]	—	—

* $p > 0.05$ = ns (not significant)

cohorts were not significantly different with regard to age, race, sex, or BMI.

ME/CFS is associated with increased prevalence of GI symptoms and irritable bowel syndrome

In comparison with healthy controls, ME/CFS participants reported a significantly higher presence of all assessed GI symptoms, including bloating ($p = 0.0001$), abdominal/stomach pain ($p < 0.0001$), bowel problems ($p < 0.0001$), nausea ($p < 0.0001$), and loss of appetite ($p < 0.0001$) (Fig. 1a). In addition, a significantly greater proportion of ME/CFS participants reported having previously received a diagnosis of IBS from a health-care professional compared with controls (53.4% vs. 7.5%, respectively; $p < 0.0001$) (Fig. 1b). Illness duration, defined as the time since ME/CFS symptom onset, was not associated with GI symptom burden scores.

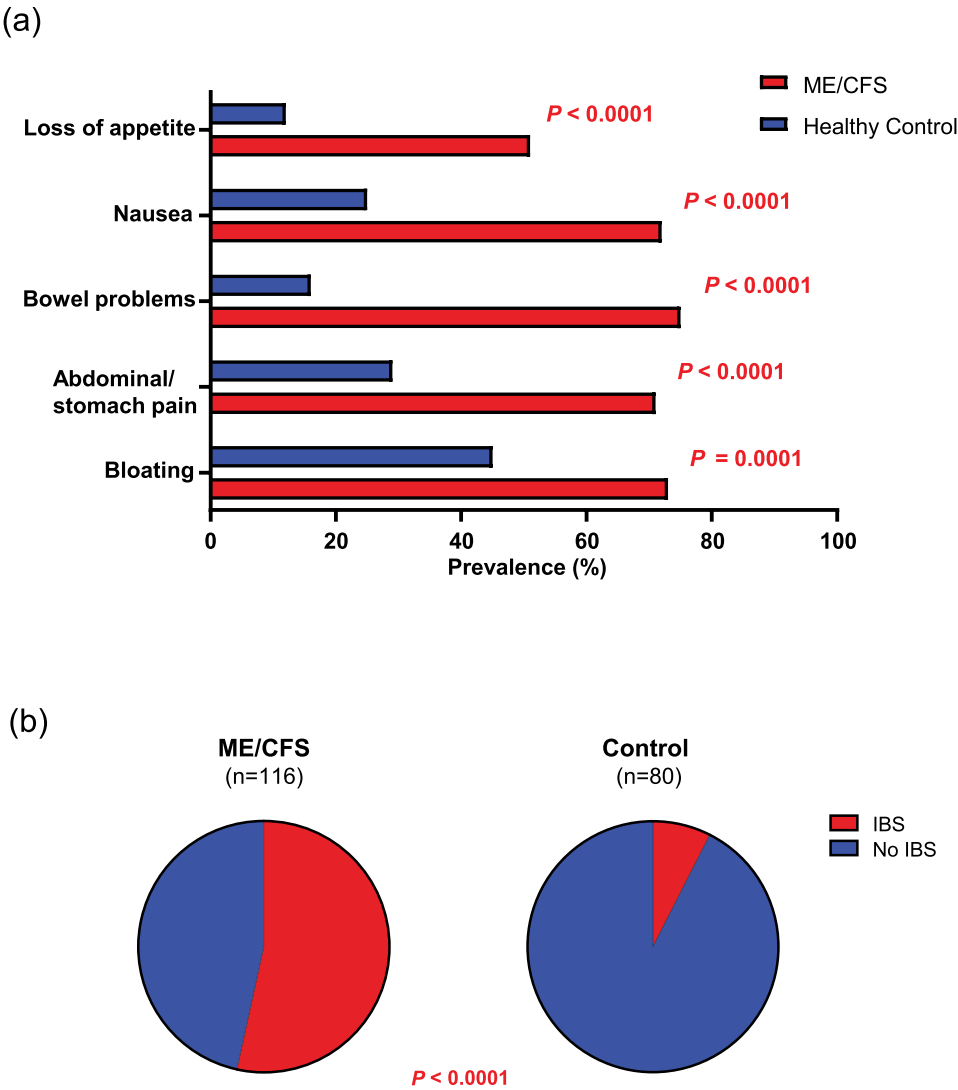


Fig. 1 GI symptoms and IBS are significantly more prevalent in ME/CFS. (a) Prevalence of individual GI symptoms and (b) IBS in ME/CFS patients ($n = 116$) and healthy controls ($n = 80$). GI symptom prevalence was defined as a frequency score ≥ 2 for each symptom

GI symptom burden correlates with core ME/CFS symptom domains

GI symptom burden scores (combined frequency and severity scores), significantly correlated with the symptom burden scores for each ME/CFS hallmark symptom

domain, including physical fatigue, cognitive difficulties and mental fatigue, musculoskeletal pain, sleep disturbances, flu-like symptoms, neurological symptoms, sensory sensitivities, and cardiovascular symptoms ($p < 0.0001$ for each; Fig. 2a–h). The strongest correlation

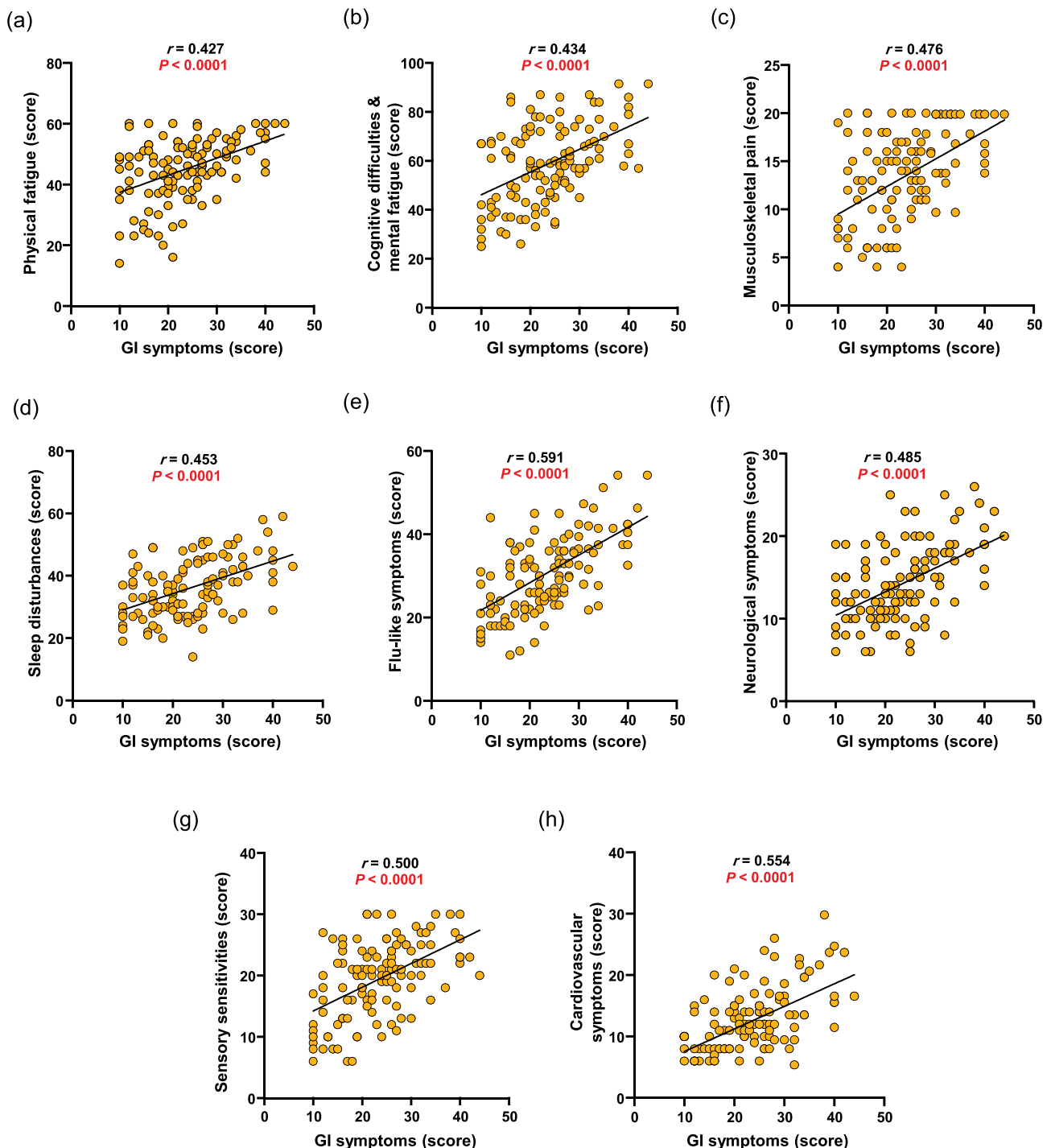


Fig. 2 GI symptom burden correlates with core ME/CFS symptom domains. (a–h) correlations between GI symptom burden scores and (a) physical fatigue, (b) cognitive difficulties and mental fatigue, (c) musculoskeletal pain, (d) sleep disturbances, (e) flu-like symptoms, (f) neurological symptoms, (g) sensory sensitivities, and (h) cardiovascular symptoms in ME/CFS patients ($n = 116$). Symptom burden scores were calculated as the sum of frequency and severity scores for each domain

Table 3 Correlations between GI and other ME/CFS symptom domain burden scores

ME/CFS Symptom Domain Correlated with GI Symptoms	Spearman's r	P value
Physical Fatigue	0.427	<0.0001
Cognitive Difficulties & Mental Fatigue	0.434	<0.0001
Musculoskeletal Pain	0.476	<0.0001
Sleep Disturbances	0.453	<0.0001
Flu-Like Symptoms	0.591	<0.0001
Neurological Symptoms	0.485	<0.0001
Sensory Sensitivities	0.500	<0.0001
Cardiovascular Symptoms	0.554	<0.0001

was with flu-like symptoms ($r=0.591$). Detailed correlation statistics are provided in Table 3. All reported correlations remained significant after false discovery rate correction (Supplementary Table 1). Similar patterns

were observed when frequency and severity scores were analyzed separately (Supplementary Figures 1–2).

GI symptoms and multiple core ME/CFS domains correlate with plasma CRP
Plasma CRP levels were significantly higher in ME/CFS patients with increased GI symptom frequency (combined GI frequency score > 10) ($p=0.002$) (Fig. 3a). There was also a modest but significant correlation between CRP levels and overall GI symptom burden within the ME/CFS cohort ($r=0.212$, $p=0.022$) (Fig. 3b). Significant associations were not observed in the healthy control group.

CRP levels further correlated with symptom burden scores in several ME/CFS domains, including musculoskeletal pain, physical fatigue, and flu-like symptoms ($r=0.429$, $p<0.0001$; $r=0.353$, $p=0.0001$; and $r=0.248$, $p=0.007$ respectively) (Fig. 3c–e). These CRP correlations

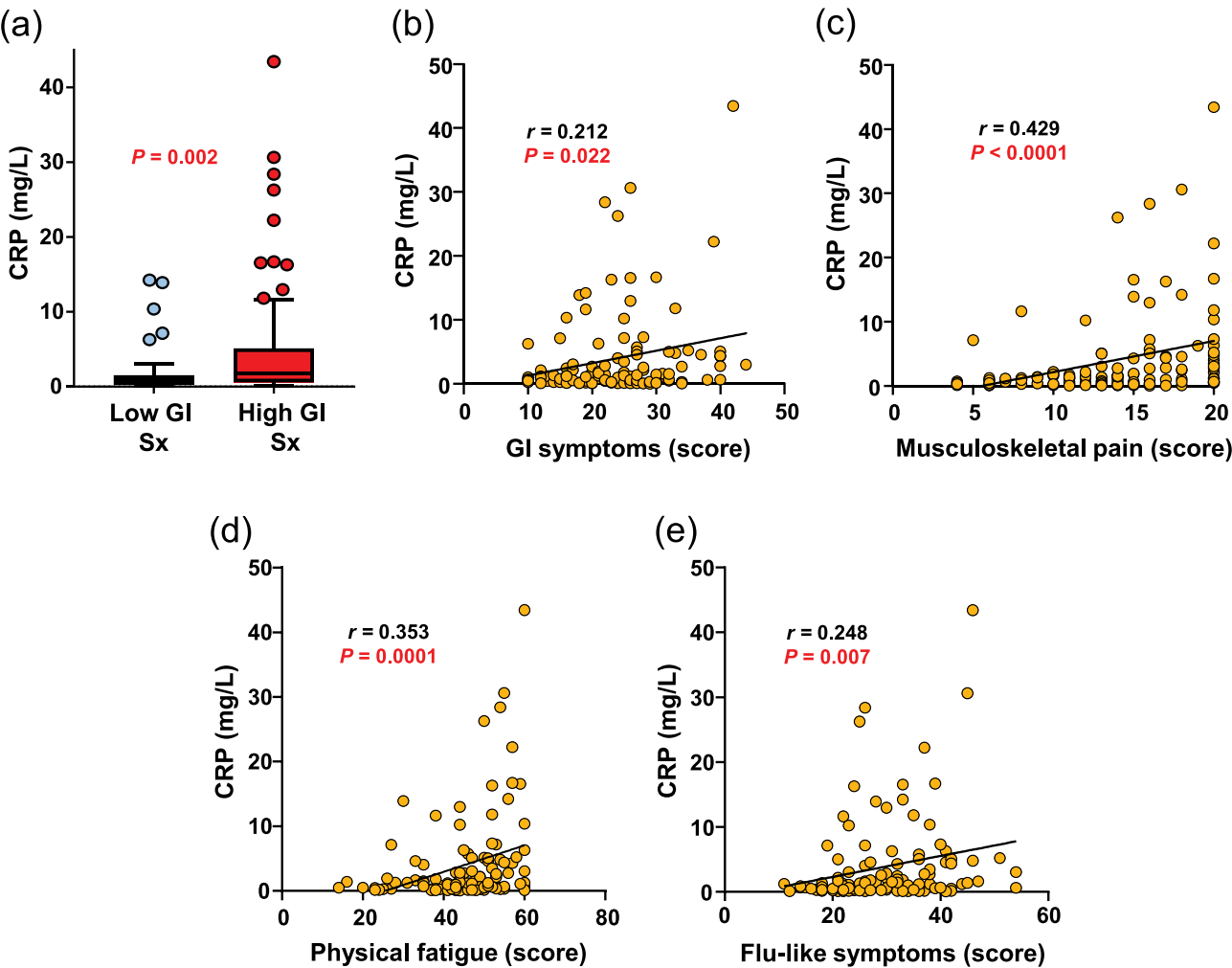


Fig. 3 GI symptoms and core ME/CFS symptom domains are associated with plasma CRP. (a) Plasma CRP levels in ME/CFS patients with high ($n=71$) versus low ($n=45$) GI symptom frequency, shown as Tukey box plots (median, IQR; whiskers = $1.5 \times$ IQR; outliers shown individually). (b–e) correlations between plasma CRP and symptom burden for GI symptoms, physical fatigue, musculoskeletal pain, and flu-like symptoms ($n=116$)

with specific symptom domains for the ME/CFS cohort remained significant after false discovery rate correction (Supplementary Table 1). Significant correlations were not observed with cognitive difficulties and mental fatigue, sleep disturbances, neurological symptoms, sensory sensitivities, or cardiovascular symptoms. No significant associations were detected in the healthy control group.

Flu-like symptoms are associated with increased IgM reactivity to dietary and microbial antigens

Associations between antigen-specific antibody levels and all predefined symptom domains were examined. Specifically, we assessed circulating levels of IgG, IgA, and IgM antibodies to microbial and dietary targets as indirect markers potentially consistent with gut-derived antigen exposure, which may result from impaired gut barrier integrity, as discussed in prior studies [20, 35]. Significant relationships were observed only for the flu-like symptom domain. Within the ME/CFS cohort, anti-gliadin IgM and anti-LPS IgM levels demonstrated modest correlations with flu-like symptom burden scores ($r=0.210$, $p=0.024$; $r=0.196$, $p=0.035$, respectively) (Fig. 4a, c). While these correlations were nominally significant at the p -value level, they did not remain significant after false discovery rate correction (Supplementary Table 1). Comparable associations were not observed for the other predefined hallmark symptom domains or in the healthy control group.

Based on these observations, patients were subsequently stratified by flu-like symptom frequency. Patients with higher flu-like symptom frequency exhibited significantly elevated anti-gliadin IgM and anti-LPS IgM levels ($p=0.022$ and $p=0.010$, respectively) (Fig. 4b,d).

Antibodies to gliadin are not associated with serologic and genetic markers of celiac disease in ME/CFS

HLA genotyping indicated no association between antibody reactivity to native gliadin and the presence of celiac disease-associated HLA-DQ2 and/or HLA-DQ8 alleles in the ME/CFS cohort. Among the 18 ME/CFS individuals with anti-gliadin IgG and/or IgA levels exceeding three standard deviations above the mean of the control group, only six were positive for HLA-DQ2 and/or HLA-DQ8, a frequency comparable to that reported in the general population [41]. In addition, no significant differences were observed between ME/CFS patients and healthy controls in the levels of established serologic markers of celiac disease, including IgA antibodies to TG2 and IgG or IgA antibodies to deamidated gliadin.

Discussion

Our findings provide evidence that gastrointestinal symptomatology is both common and clinically relevant in ME/CFS, underscoring its importance as a core dimension of the illness rather than an ancillary complaint. Patients reported markedly elevated rates of bloating, abdominal pain, nausea, appetite loss, and bowel disturbances, with more than half reporting a prior diagnosis of IBS. While GI involvement has been noted previously, prior studies have addressed it in isolation or in smaller cohorts. By integrating detailed symptom profiling with hallmark ME/CFS domains and specific immunological markers, this study offers a more comprehensive view of GI dysfunction within the broader clinical and biological landscape of ME/CFS.

Among the most notable findings was the breadth of associations between GI symptom burden and nearly all major ME/CFS domains, including fatigue, cognitive and mental difficulties, musculoskeletal pain, sleep disturbances, flu-like symptoms, neurological complaints, sensory sensitivities, and cardiovascular manifestations. The moderate to strong correlations across domains suggest that GI symptoms represent an important component of the broader multisystem involvement in ME/CFS, rather than a separate or peripheral feature of the illness. Notably, the absence of correlation between GI symptom burden and illness duration suggests that GI involvement may persist throughout the course of illness.

At the immunological level, greater flu-like symptom burden was nominally associated with higher plasma levels of IgM antibodies against dietary and microbial antigens. This isotype-specific association, in the absence of corresponding IgG or IgA relationships, may suggest ongoing or recurrent antigenic stimulation rather than a generalized increase in antibody production. Elevations in these markers, which were not observed in controls, are directionally consistent with increased gut-derived antigen exposure and heightened immune response in a subset of patients, but should be considered exploratory and require confirmation in independent cohorts. The pattern suggests that flu-like symptom expression may, in part, be linked to systemic immune responses triggered by increased exposure to gut-derived immunogenic factors. These findings support and extend prior work implicating gut-associated antigen exposure and associated immune alterations in ME/CFS [20, 42], providing additional serological evidence for gut-immune axis disruption as a potential upstream pathophysiological event.

In parallel with these observations, plasma CRP levels were significantly higher in patients with greater GI symptom burden and additionally correlated with fatigue, musculoskeletal pain, and flu-like features. Importantly, although prior work in the same cohort did not demonstrate overall CRP elevation in ME/CFS

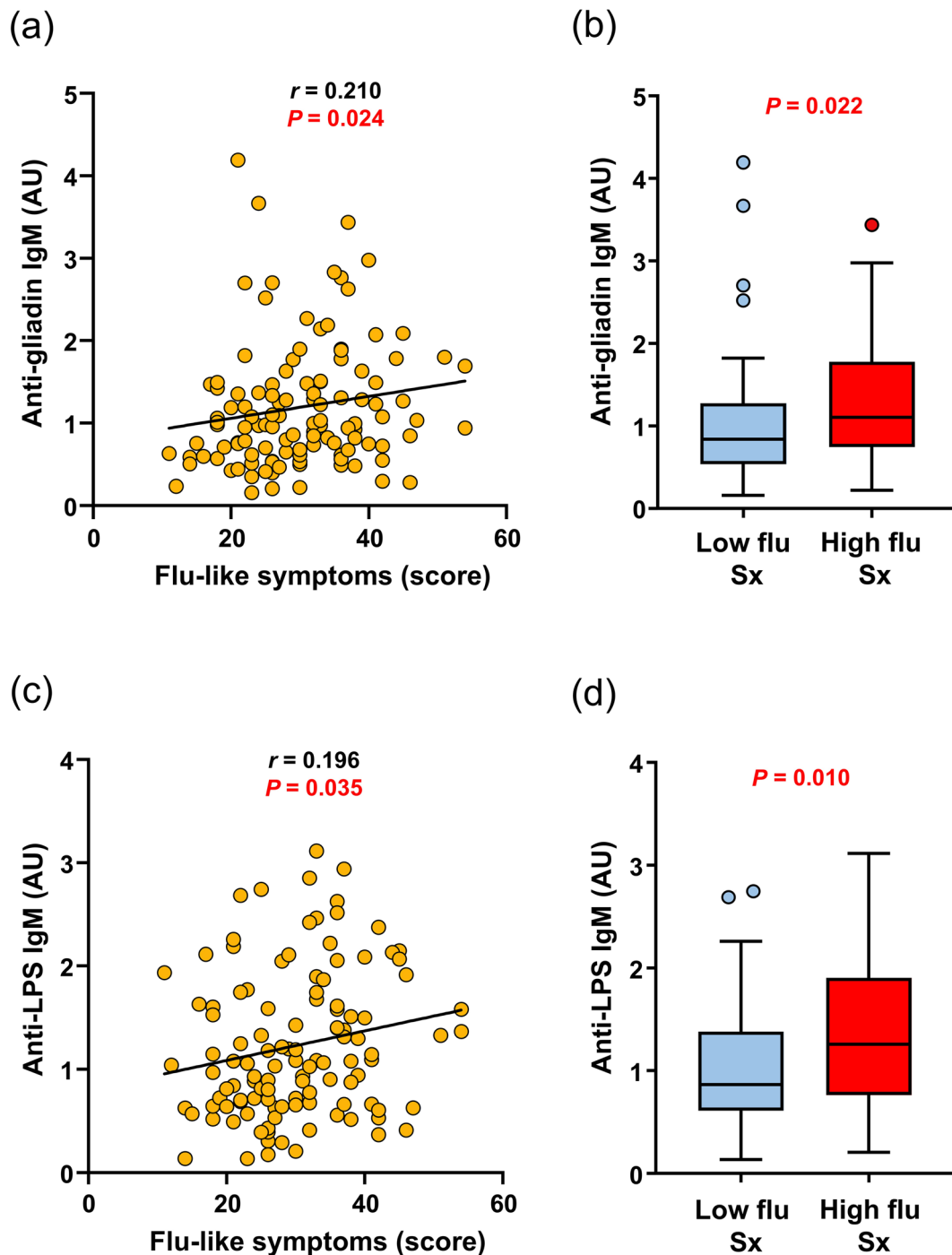


Fig. 4 Flu-like symptoms are associated with elevated IgM antibodies to dietary and microbial antigens. **(a, c)** correlations between flu-like symptom burden and anti-gliadin and anti-LPS IgM levels in ME/CFS patients ($n = 116$). **(b, d)** anti-gliadin and anti-LPS IgM levels in ME/CFS patients with high ($n = 70$) versus low ($n = 46$) flu-like symptom frequency, shown as Tukey box plots

relative to controls [22], the present symptom-stratified analyses highlight disease heterogeneity and identify a subgroup characterized by low-grade systemic inflammation that may be linked to GI disturbance. This pattern is consistent with modest immune activation and suggests that gut-derived antigen exposure could contribute to

sustained inflammatory activity in a subset of individuals with ME/CFS.

From a mechanistic perspective, the convergence of clinical and immunological findings in this study is compatible with the hypothesis that GI dysfunction may contribute to the expression or amplification of core ME/CFS

symptoms through multiple, interrelated biological pathways. Mounting evidence from ME/CFS cohorts indicates that intestinal dysbiosis, characterized by reduced microbial diversity and altered metabolic output, can potentiate inflammatory signaling [18–20, 43]. Disruption of gut barrier function can lead to the translocation of microbial products into systemic circulation, driving immune activation, low-grade inflammation, and the release of pro-inflammatory cytokines that can influence fatigue, pain, cognitive function, and mood [44–48]. The strong correlation of GI burden with multiple core symptom domains further reinforces this model. In this context, the GI tract may represent both a site of involvement and a contributor to symptom variability and persistence. However, causal relationships will require confirmation through prospective and interventional studies.

Therapeutically, these findings provide a biologically plausible rationale for further exploration of gut-immune targeted interventions in ME/CFS, particularly among individuals with pronounced GI symptom burden. Although clinical investigations of microbiome-directed approaches, including probiotics, dietary interventions, and fecal microbiota transplantation, remain at an early stage and have yielded variable results, the heterogeneity in GI involvement observed here underscores the potential value of symptom-based stratification in both research and clinical contexts [49–51]. Careful subgrouping by GI phenotype may help identify patient populations more likely to benefit, potentially improving trial efficiency and supporting the development of more individualized therapeutic strategies. Additional well-controlled studies are needed to determine whether targeting the gut ecosystem, barrier integrity, or downstream immune consequences can translate into clinically meaningful benefits in ME/CFS and to clarify which patient subsets are most responsive.

This study has several strengths, including a comparatively well-powered cohort for ME/CFS research, comprehensive symptom phenotyping, and integration of patient-reported outcomes with immunological measures. Nevertheless, important limitations warrant consideration. First, although the sample size is substantial relative to prior ME/CFS studies, larger and more clinically diverse cohorts will be necessary to further validate subgroup-specific findings and improve generalizability. Second, GI and ME/CFS symptom assessments relied on self-reported questionnaire data and may therefore be subject to recall and reporting biases. Although standardized patient-reported symptom instruments are widely used in ME/CFS research, future studies incorporating objective physiological, cognitive, autonomic, GI, and sleep-related measures will be important for strengthening and extending these findings. Third, biomarker analyses were limited to a targeted serological panel, and

additional direct measures relevant to gut barrier function, such as lactulose-mannitol testing, were not available in this cohort, limiting broader characterization of GI, immunologic, and metabolic alterations associated with ME/CFS. Detailed information regarding GI-directed medications, probiotics, dietary interventions, and anti-inflammatory therapies was also not systematically available and may represent potential confounding factors. Finally, the cross-sectional design prevents inferences regarding causality or temporal sequence. Future studies incorporating expanded immune profiling, microbiome and metabolomic analyses, and, when feasible, longitudinal or interventional designs, particularly those capturing dynamic responses to physiological stressors such as exertion, will be important for clarifying the mechanistic relationships between GI dysfunction, immune activation, and symptom expression in ME/CFS.

In conclusion, this study demonstrates that GI symptom burden is a prominent and clinically relevant feature of ME/CFS that aligns with broader symptom patterns and specific indicators of systemic immune activation. The combined observations of domain-level symptom correlations, flu-like symptom associations with antigen-specific IgM responses, and stratified elevations in plasma CRP support the involvement of gut-related immune processes in a subset of patients and highlight inflammatory heterogeneity within the condition. Recognizing GI involvement as part of the multisystem presentation of ME/CFS may help refine symptom-based stratification approaches and guide future investigations aimed at identifying targeted, mechanism-informed interventions.

Abbreviations

ME/CFS	myalgic encephalomyelitis/chronic fatigue syndrome
GI	gastrointestinal
IBS	irritable bowel syndrome
IBD	inflammatory bowel disease
CRP	C-reactive protein
LPS	lipopolysaccharide
ELISA	enzyme-linked immunosorbent assay

Supplementary information

The online version contains supplementary material available at <https://doi.org/10.1186/s12967-026-08442-1>.

Supplementary Material 1

Supplementary Material 2

Supplementary Material 3

Supplementary Material 4

Authors' contributions

Study concept and design: MB, SDV, PHG, AA; Acquisition of data: ACI, SDV; Analysis and/or interpretation of data: MB, SDV, PHG, AA; Drafting of the manuscript: MB, AA; Critical revision of the manuscript for important intellectual content: MB, SDV, ACI, PHG, AA; Statistical analysis: MB, AA;

Administrative, technical, or material support: SDV, PHG, AA; Obtained funding: AA; Study supervision: AA.

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Data availability

The dataset supporting the conclusions of this article are included within the article (and its additional files). Data will be made available on reasonable request.

Declarations

Ethical approval and consent to participate

This study was approved by the institutional review board of Columbia University Medical Center and was conducted in accordance with the Declaration of Helsinki.

Consent for publication

Not applicable.

Competing interests

AA reports participation on advisory panels for the National Institutes of Health, Department of Defense, and Global Lyme Alliance. Other authors report no relevant disclosures.

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